

Python REST API Assignment

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Step 1: Installed Sqlite3 using chocolaty: `choco install sqlite -y`

```
(base) PS C:\WINDOWS\system32> choco -v
2.5.1
(base) PS C:\WINDOWS\system32> choco install sqlite -y
Chocolatey v2.5.1
Installing the following packages:
sqlite
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading chocolatey-compatibility.extension 1.0.0... 100%

chocolatey-compatibility.extension v1.0.0 [Approved]
chocolatey-compatibility.extension package files install completed. Performing other installation steps.
Installed/updated chocolatey-compatibility extensions.
The install of chocolatey-compatibility.extension was successful.
Deployed to 'C:\ProgramData\chocolatey\extensions\chocolatey-compatibility'
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading chocolatey-core.extension 1.4.0... 100%

chocolatey-core.extension v1.4.0 [Approved]
chocolatey-core.extension package files install completed. Performing other installation steps.
Installed/updated chocolatey-core extensions.
The install of chocolatey-core.extension was successful.
Deployed to 'C:\ProgramData\chocolatey\extensions\chocolatey-core'
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading SQLite 3.50.2... 100%
```

Checking version: `sqlite3 --version`

```
(base) PS C:\WINDOWS\system32> sqlite3 --version
3.45.3 2024-04-15 13:34:05 8653b758870e6ef0c98d46b3ace27849054af85da891eb121e9aaa537f1e8355 (64-bit)
(base) PS C:\WINDOWS\system32>
```

Checking our database is working or not: `sqlite3 northwind.db`

```
(base) PS F:\Internship\Python\rest_api> sqlite3 northwind.db
SQLite version 3.45.3 2024-04-15 13:34:05 (UTF-16 console I/O)
Enter ".help" for usage hints.
sqlite> .tables
Alphabetical list of products      Orders Qty
Categories                        Product Sales for 1997
Category Sales for 1997           ProductDetails_V
Current Product List              Products
Customer and Suppliers by City    Products Above Average Price
Invoices                          Shippers
Order Details                     Summary of Sales by Quarter
Order Details Extended            Summary of Sales by Year
Order Subtotals                    Suppliers
Orders                            Territories
sqlite> SELECT * FROM Customers LIMIT 5;
ALFKI|Alfreds Futterkiste|Maria Anders|Sales Representative|Obere Str. 57|Berlin|Western Europe|12209|Germany|030-0074321|030-0076545
ANATR|Ana Trujillo Emparedados y helados|Ana Trujillo|Owner|Avda. de la Constitución 2222|México D.F.|Central America|05021|Mexico|(5) 555-4729|(5) 555-3745
ANTON|Antonio Moreno Taquería|Antonio Moreno|Owner|Mataderos  2312|México D.F.|Central America|05023|Mexico|(5) 555-3932|
AROUT|Around the Horn|Thomas Hardy|Sales Representative|120 Hanover Sq.|London|British Isles|WA1 1DP|UK|(171) 555-7788|(171) 555-6750
BERGS|Berglunds snabbköp|Christina Berglund|Order Administrator|Berguvsvägen  8|Luleå|Northern Europe|S-958 22|Sweden|0921-12 34 65|0921-12 34 67
sqlite> 
```

Note: In **GET /customers?limit=10 offset=0**, the **offset=0** parameter specifies that zero rows should be skipped before the results are returned. It indicates that the result set should begin from the very first row of data, making it equivalent to **offset=0**. This setup is commonly used in pagination to display the first page of results when a larger dataset is being retrieved.

Step 2: Created the folder structure:

```
northwind-rest-api/  
├── app.py  
├── config.py  
├── requirements.txt  
  
├── controllers/  
│   ├── __init__.py  
│   ├── customers.py  
│   ├── orders.py  
│   └── products.py  
  
├── models/  
│   ├── __init__.py  
│   ├── customer.py  
│   ├── order.py  
│   └── product.py  
  
├── schemas/  
│   ├── __init__.py  
│   ├── customer_schema.py  
│   ├── order_schema.py  
│   └── product_schema.py  
  
├── tests/  
│   ├── conftest.py  
│   ├── test_customer.py  
│   ├── test_order.py  
│   └── test_product.py  
  
└── northwind.db
```

Step 3: Installed all packages from requirements.txt using:

```
pip install -r requirements.txt
```

Step 3: Checked normally whether all API endpoints are working correctly or not:

```
@app.route("/customers/<customer_id>", methods=["GET"])
def get_customer(customer_id):
    """GET /customers/<CustomerID>"""
    db = get_db()
    cur = db.execute(
        """
        SELECT CustomerID, CompanyName, ContactName, ContactTitle,
           Address, City, Region, PostalCode, Country, Phone, Fax
        FROM Customers
        WHERE CustomerID = ?
        """,
        (customer_id,),
    )
    row = cur.fetchone()
    if row is None:
        return jsonify({"error": "Customer not found"}), 404
    return jsonify(dict(row)), 200
```

The screenshot shows a REST client interface with the following details:

- URL:** `{{base_url}}/customers/ALFKI`
- Method:** GET
- Params:** Authorization, Headers (6), Body, Scripts, Settings
- Query Params:** A table with 4 columns: Key, Value, Description. The first row has 'Key' in the first column and 'Value' in the second column.
- Body:** JSON format. The response is a 200 OK status with a 25 ms response time.
- JSON Response:**

```
{
  "Address": "Obere Str. 57",
  "City": "Berlin",
  "CompanyName": "Alfreds Futterkiste",
  "ContactName": "Maria Anders",
  "ContactTitle": "Sales Representative",
  "Country": "Germany",
  "CustomerID": "ALFKI",
  "Fax": "030-0076545",
  "Phone": "030-0074321",
  "PostalCode": "12209",
  "Region": "Western Europe"
}
```

Got 200 status code...

- Deleted the customer "ALFKI ".

HTTP

{{base_url}}/customers/ALFKI

Save

DELETE

{{base_url}}/customers/ALFKI

Params

Authorization

Headers (6)

Body

Scripts

Settings

Query Params

	Key	Value	Description
	Key	Value	Description

Body

Cookies

Headers (5)

Test Results

200 OK • 124 ms

{ } JSON

Preview

Visualize

1 {

2 | "message": "Customer ALFKI deleted"

3 }

- No trace found for the customer "ALFKI " after deletion.

HTTP

{{base_url}}/customers/ALFKI

Save

GET

{{base_url}}/customers/ALFKI

Params

Authorization

Headers (6)

Body

Scripts

Settings

Query Params

☐	Key	Value	Description
☐			
	Key	Value	Description

Body

Cookies

Headers (5)

Test Results

404 NOT FOUND • 29 ms

{ } JSON

Preview

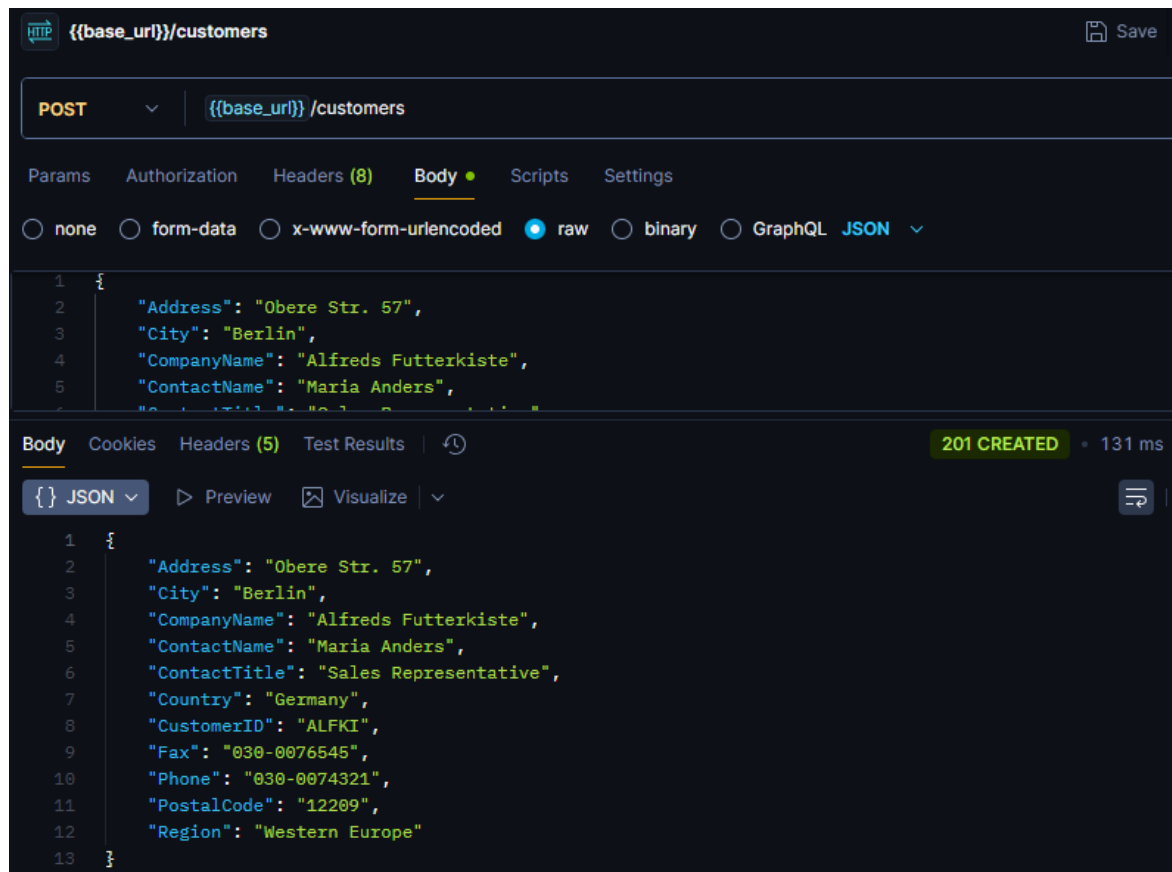
Debug with AI

1 {

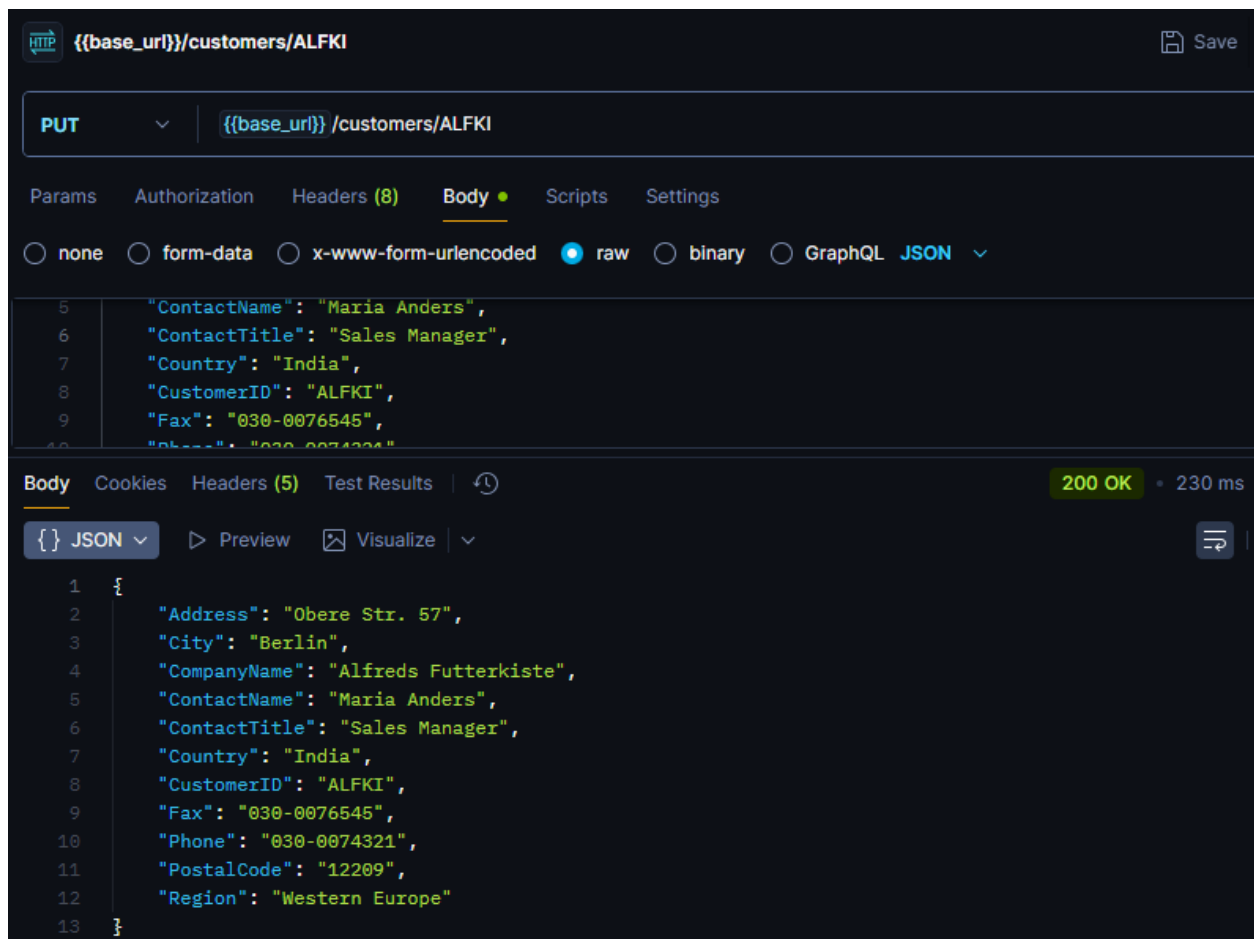
2 | "error": "Customer not found"

3 }

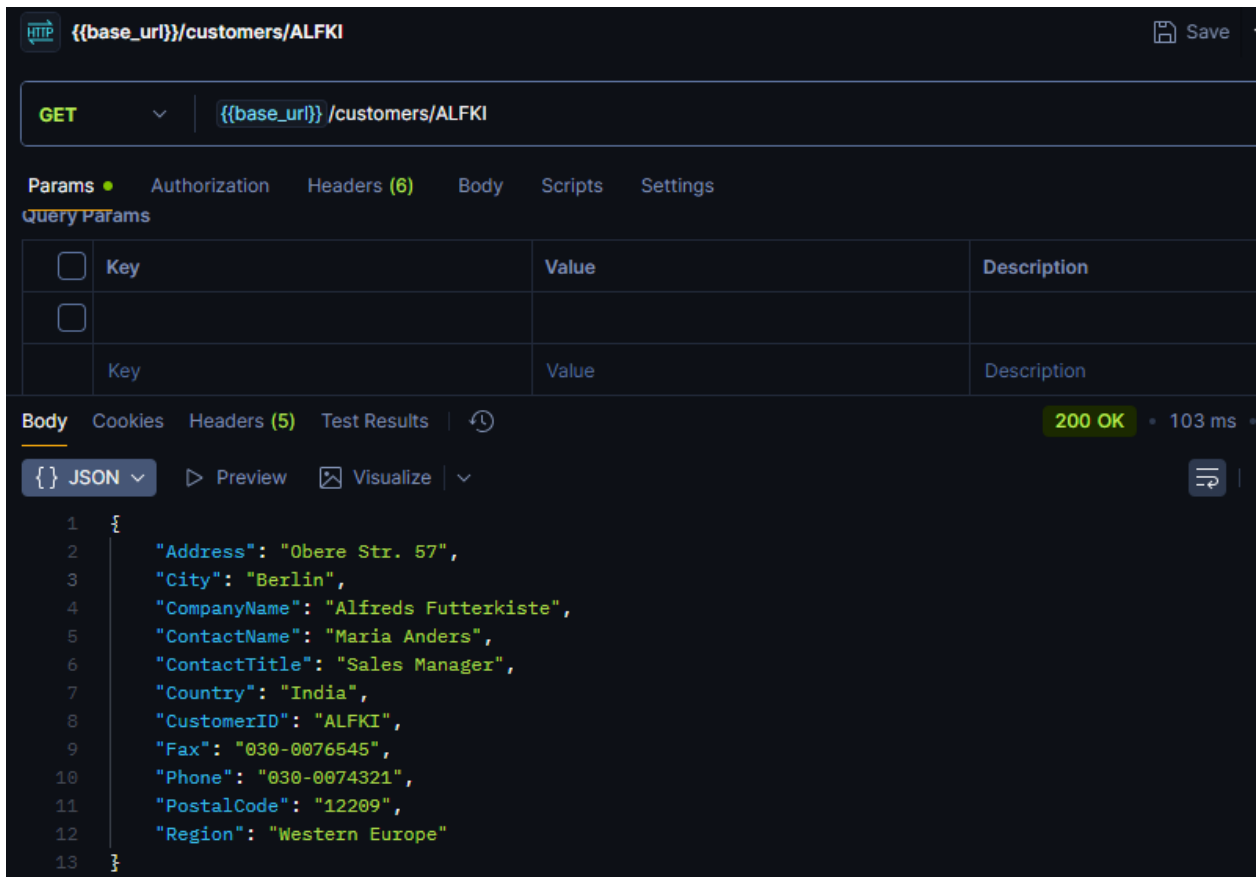
- Created a new customer "ALFKI" again, by giving raw JSON input in body section.



- We have updated the "ContactTitle": "Sales Representative" to "ContactTitle": "Sales Manager" and "Country": "Germany" to "Country": "India" successfully.



- Now if we again search for the customer "ALFKI", we get the updated details.



HTTP {{base_url}}/customers/ALFKI

GET {{base_url}}/customers/ALFKI

Params • Authorization Headers (6) Body Scripts Settings

Query Params

Key	Value	Description
Key	Value	Description

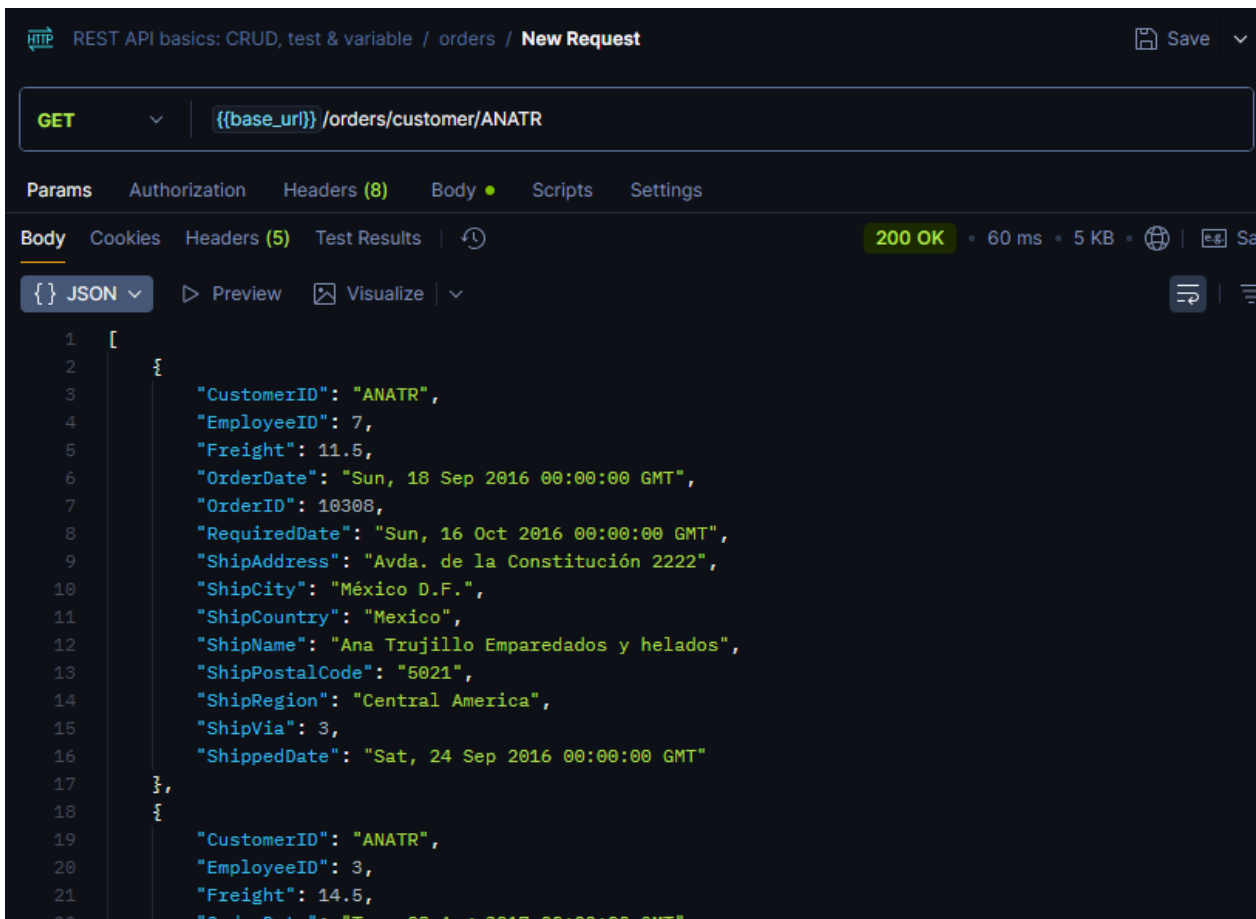
Body Cookies Headers (5) Test Results 200 OK • 103 ms

{ JSON Preview Visualize

```

1 {
2   "Address": "Obere Str. 57",
3   "City": "Berlin",
4   "CompanyName": "Alfreds Futterkiste",
5   "ContactName": "Maria Anders",
6   "ContactTitle": "Sales Manager",
7   "Country": "India",
8   "CustomerID": "ALFKI",
9   "Fax": "030-0076545",
10  "Phone": "030-0074321",
11  "PostalCode": "12209",
12  "Region": "Western Europe"
13 }
```

- So finally, all our endpoints are working 100% fine for all products, orders and customers. Also, we have implemented the order history for a given customer.



REST API basics: CRUD, test & variable / orders / New Request

GET {{base_url}}/orders/customer/ANATR

Params Authorization Headers (8) Body • Scripts Settings

Body Cookies Headers (5) Test Results 200 OK • 60 ms • 5 KB

{ JSON Preview Visualize

```

1 [
2   {
3     "CustomerID": "ANATR",
4     "EmployeeID": 7,
5     "Freight": 11.5,
6     "OrderDate": "Sun, 18 Sep 2016 00:00:00 GMT",
7     "OrderID": 10308,
8     "RequiredDate": "Sun, 16 Oct 2016 00:00:00 GMT",
9     "ShipAddress": "Avda. de la Constitución 2222",
10    "ShipCity": "México D.F.",
11    "ShipCountry": "Mexico",
12    "ShipName": "Ana Trujillo Emparedados y helados",
13    "ShipPostalCode": "5021",
14    "ShipRegion": "Central America",
15    "ShipVia": 3,
16    "ShippedDate": "Sat, 24 Sep 2016 00:00:00 GMT"
17  },
18  {
19    "CustomerID": "ANATR",
20    "EmployeeID": 3,
21    "Freight": 14.5,
22    "OrderDate": "Tue, 08 Aug 2017 00:00:00 GMT"
```

Step 4: Tried unit testing on products, customers and orders:

```
===== warnings summary =====
tests/test_product.py::test_get_product
  F:\Internship\Python\rest_api\northwind-rest-api\controllers\products.py:15: LegacyAPIWarning: The Query.get() method
  SQLAlchemy and becomes a legacy construct in 2.0. The method is now available as Session.get() (deprecated since: 2.0) (
  he.me/e/b8d9)
    p = Product.query.get(product_id)

tests/test_product.py::test_create_product
  if Product.query.get(schema.ProductID):

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== short test summary info =====
ERROR tests/test_product.py::test_update_product - sqlalchemy.exc.IntegrityError: (sqlite3.IntegrityError) UNIQUE ...
ERROR tests/test_product.py::test_delete_product - sqlalchemy.exc.PendingRollbackError: This Session's transaction...
===== 13 passed, 2 warnings, 2 errors in 1.75s =====
(testing_DG_1) PS F:\Internship\Python\rest_api\northwind-rest-api>
```

- After removing errors, warnings, bugs and correcting the code:

```
(testing_DG_1) PS F:\Internship\Python\rest_api\northwind-rest-api> python -m pytest
===== test session starts =====
platform win32 -- Python 3.13.5, pytest-8.4.2, pluggy-1.5.0
rootdir: F:\Internship\Python\rest_api\northwind-rest-api
plugins: anyio-4.7.0, cov-7.0.0
collected 15 items

tests\test_customer.py ..... [ 33%]
tests\test_order.py ..... [ 66%]
tests\test_product.py ..... [100%]

===== 15 passed in 0.64s =====
(testing_DG_1) PS F:\Internship\Python\rest_api\northwind-rest-api>
```

Step 5: Generating code coverage report:

```
(testing_DG_1) PS F:\Internship\Python\rest_api\northwind-rest-api> python -m pytest --cov=config --cov=controllers --cov=models --cov=schema --cov-report=term-missing
----- coverage: platform win32, python 3.13.5-final-0 -----
Name                               Stmts  Miss  Cover   Missing
-----
config.py                           7      0   100%
controllers\__init__.py             0      0   100%
controllers\customers.py           48      0   100%
controllers\orders.py              55     13    76%   24-25, 28, 38, 44-45, 55, 61-67
controllers\products.py            48      0   100%
models\__init__.py                  0      0   100%
models\customer.py                 16      0   100%
models\order.py                    19      0   100%
models\product.py                  15      0   100%
-----
TOTAL                               208     13    94%
===== 15 passed in 0.79s =====
C:\Users\Admin\anaconda3\Lib\site-packages\_pytest\unraisableexception.py:33: ResourceWarning: unclosed database in <sqlite3.Connection object at 0x000002003ABC8220>
  gc.collect()
ResourceWarning: Enable tracemalloc to get the object allocation traceback
(testing_DG_1) PS F:\Internship\Python\rest_api\northwind-rest-api>
```

- After finding all the missing and unreachable codes, we corrected it:’

```
(testing_DG_1) PS F:\Internship\Python\rest_api\northwind-rest-api> python -m pytest --cov=config --cov=controllers --cov=models --cov=schema --cov-report=term-missing
----- coverage: platform win32, python 3.13.5-final-0 -----
Name                               Stmts  Miss  Cover   Missing
-----
config.py                           7      0   100%
controllers\__init__.py             0      0   100%
controllers\customers.py           48      0   100%
controllers\orders.py              55      0   100%
controllers\products.py            48      0   100%
models\__init__.py                  0      0   100%
models\customer.py                 16      0   100%
models\order.py                    19      0   100%
models\product.py                  15      0   100%
-----
TOTAL                               208      0   100%
===== 16 passed in 1.43s =====
C:\Users\Admin\anaconda3\Lib\site-packages\_pytest\unraisableexception.py:33: ResourceWarning: unclosed database in <sqlite3.Connection object at 0x0000016329D8C220>
  gc.collect()
ResourceWarning: Enable tracemalloc to get the object allocation traceback
(testing_DG_1) PS F:\Internship\Python\rest_api\northwind-rest-api>
```

Hence ensured, all endpoints are working correctly and checked efficiently using proper industry norms, MVC model (with appropriate folder structure) and ORM.

Summary:

- **controllers/** → All Flask route logic (customers.py, orders.py, products.py).
- **models/** → SQLAlchemy ORM models (customer.py, order.py, product.py).
- **schemas/** → Pydantic schemas for validation.
- **tests/** → pytest test cases + fixtures (conftest.py, test_customer.py, test_order.py, test_product.py).
- **northwind.db** → SQLite database (prod reference, tests use their own).
- **config.py** → DB + app setup.
- **app.py** → Flask entry point.

This project involves building a RESTful API in Python using Flask to interact with the northwind dataset. The API supports creating, retrieving, updating and deleting customers, products, and orders, along with fetching the complete order history of a specific customer. It follows the **MVC architecture** with clear separation of concerns, using SQLAlchemy for ORM and pydantic for schema validation. The codebase adheres to industry best practices, including proper error handling, input validation, and modular folder structure. Development was done using a popular IDE (VS Code) with **Git** integration for version control. Comprehensive **unit tests** were written using pytest, covering all endpoints with mocking for external service interactions.